TOPSTECHNOLOGIES
Training | Outsourcing | Placement | Study Abroad

Comptia N+ Assignment

[N+ Network Fundamental and building networks]

Configure IPv6 Addressing

Task:

❖ **1. Open your Windows or Linux system, use the command line to assign a static IPv6 address to your primary network adapter, and verify the configuration using ipconfig (Windows) or ifconfig/ip (Linux).**

➢ **Aim**

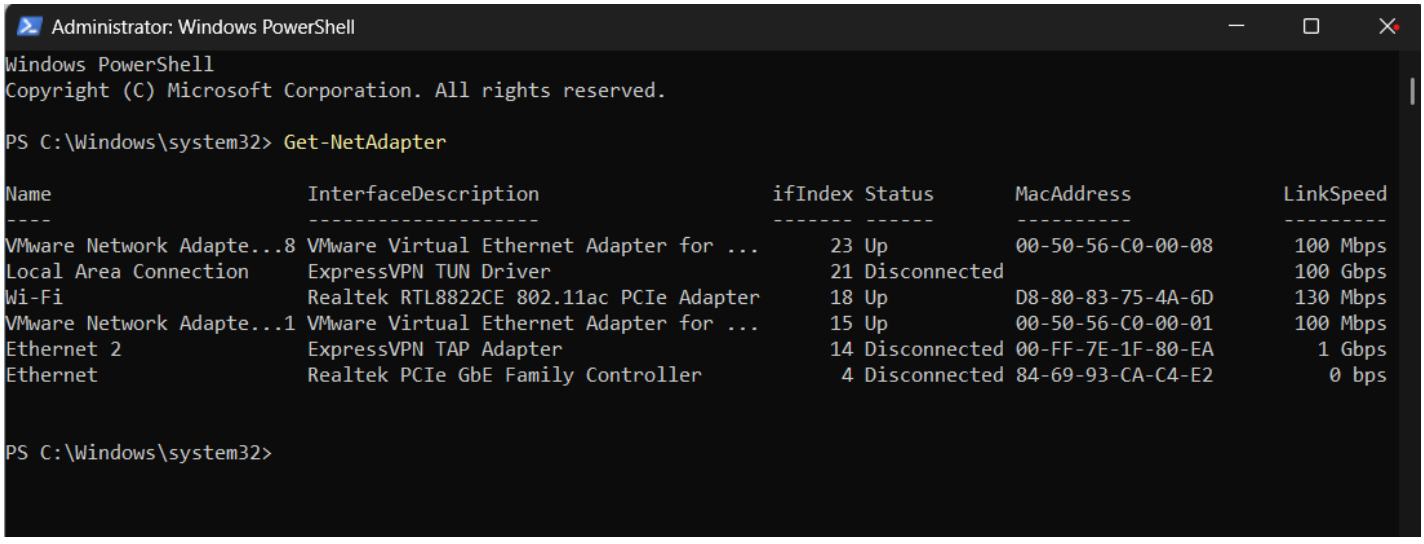
- To configure a static IPv6 address on a network adapter using the command line and verify the configuration using ipconfig on Windows or ip on Linux.

➢ **Introduction**

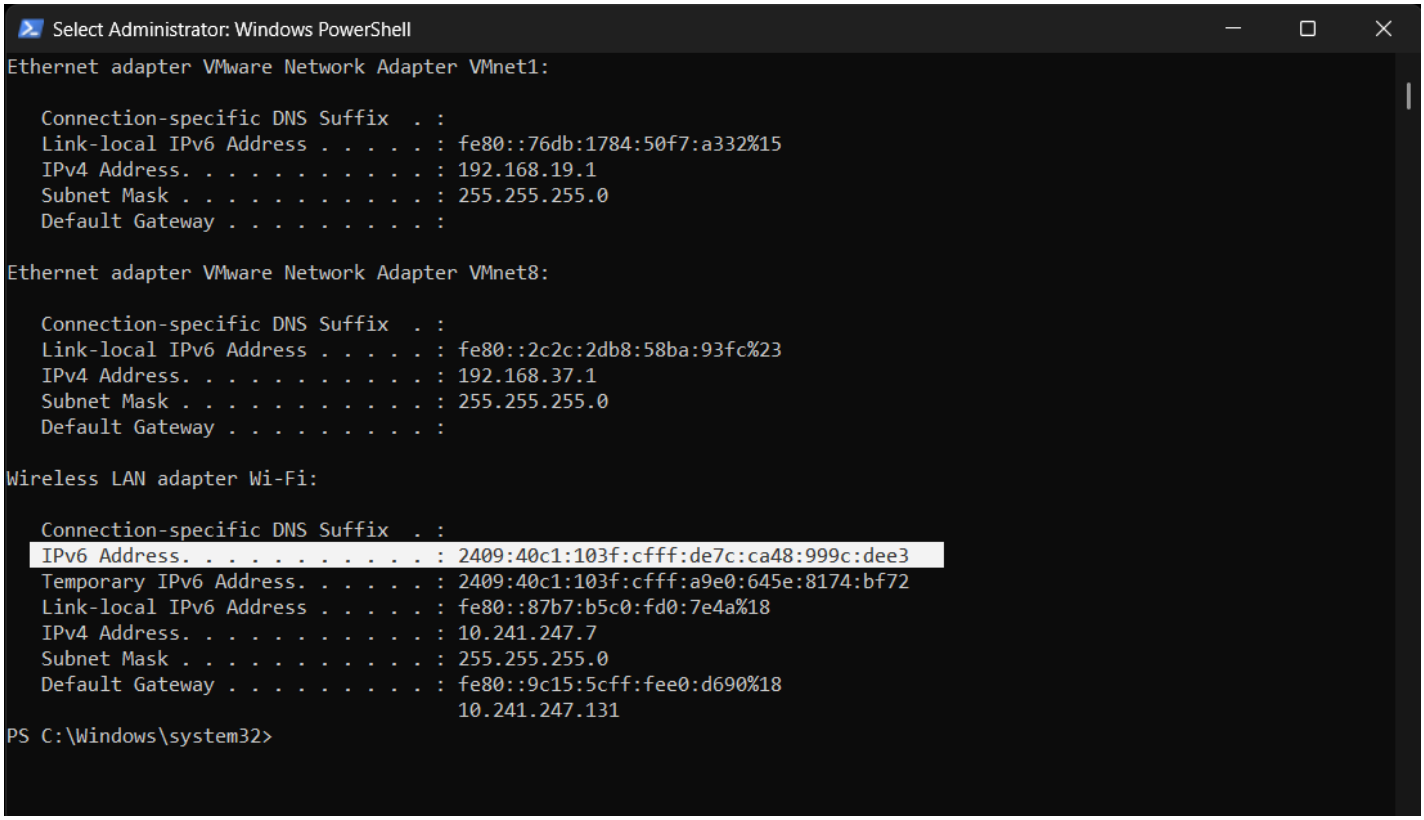
- **IPv6 (Internet Protocol Version 6)** is the latest version of the Internet Protocol used to identify devices on a network. It was developed to provide a much larger address space than IPv4.
- An IPv6 address consists of 128 bits and is written in hexadecimal notation, separated by colons. A **static IPv6 address** is manually configured and remains assigned until it is changed or removed.

- **Requirements**
  - A Windows or Linux computer
  - Administrator or root privileges
  - An active network adapter
  - A valid IPv6 address and prefix length for the test network
  - Command Prompt or Terminal

- **Procedure for Windows**
- **Step 1: Open Command Prompt or PowerShell**
  Open Windows Terminal or PowerShell as Administrator.
- **Step 2: Identify the Network Adapter**
  Execute the following command:
  Get-NetAdapter
  This command displays the available network adapters and their names, such as Ethernet or Wi-Fi.



- **Step 3: Configure a Static IPv6 Address**
- Use the following command after replacing the adapter name and example address with values valid for your network:
  New-NetIPAddress -InterfaceAlias "Ethernet" -AddressFamily IPv6 -IPAddress "2409:40c1:103f:cfff:de7c:ca48:999:dee3" -PrefixLength 64
- For example, a test network might use an address from the documentation prefix 2001:db8::/32, such as 2001:db8:1::10. This is an example only and is not a real Internet-routable address.



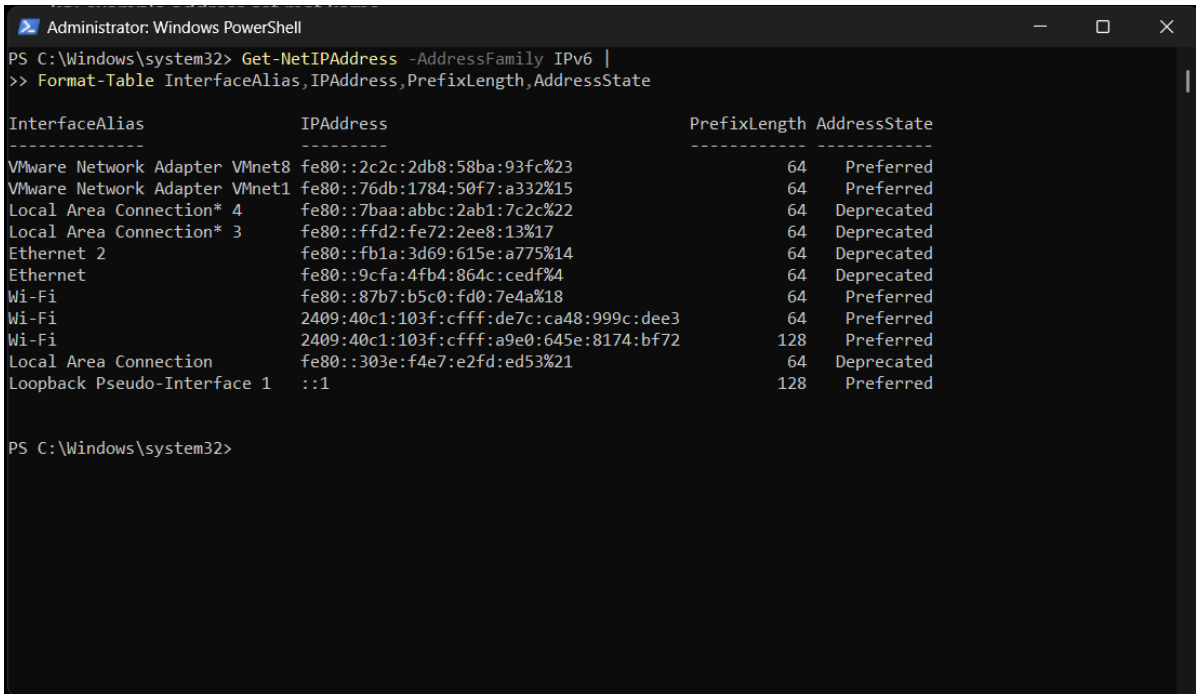
- **Step 4: Verify the IPv6 Configuration**
  Execute:

ipconfig

To display detailed IPv6 configuration, use:

ipconfig /all

- Look for the relevant network adapter and check its IPv6 Address field.



- **Procedure for Linux**
- **Step 1: Open the Terminal**

  Open a terminal window on the Linux system.
- **Step 2: Identify the Network Interface**

  Execute:

  ip link show
- This command displays the available network interfaces, such as eth0, ens33, or wlan0.
- **Step 3: Configure a Static IPv6 Address**

- Use the following command, replacing the interface name and example address with values valid for your test network:
  sudo ip -6 addr add For example, on a test network with an appropriate IPv6 prefix:
  sudo ip -6 addr add 2001:db8:1::10/64 dev eth0
- This example uses a documentation-only IPv6 prefix. It should be used only in an isolated lab or demonstration environment, not as a real Internet address.
- **Step 4: Verify the Configuration**
- Execute:
  ip -6 addr show
- Alternatively, if ifconfig is installed and supports IPv6, execute:
  ifconfig
- Look for the relevant interface and check the inet6 address entry.
- **Observation**
- After configuring the IPv6 address, the network adapter's IPv6 configuration can be examined using the appropriate verification command.

Operating System	Configuration Method	Verification Command
Windows	PowerShell networking commands	Ipconfig or ipconfig /all
Linux	ip command	`ip -6 addr show`
Linux (alternative)	`ifconfig`, if available	`ifconfig`

- The actual IPv6 address and interface name depend on the computer and the network configuration.

- **Result**
  - A static IPv6 address can be configured on a network adapter using command-line tools. The configuration can then be verified using ipconfig on Windows or ip -6 addr show on Linux.

❖ **2. Configure a unique local IPv6 address (starting with 'fd') on your system and confirm connectivity by pinging another device on your LAN that also has a unique local IPv6 address. Use the ping6 or ping -6 command for IPv6 testing.**

- **Aim**
  - To configure a Unique Local IPv6 Address (ULA) beginning with fd on a network device and verify connectivity by pinging another device on the same LAN using the ping -6 or ping6 command.

- **Introduction**
  - IPv6 (Internet Protocol Version 6) is a network-layer protocol used to identify devices and enable communication over a network.
  - A **Unique Local IPv6 Address (ULA)** is an IPv6 address designed for communication within private networks. It is similar in purpose to private IPv4 addresses such as 192.168.1.10.

- Unique Local IPv6 addresses belong to the range fc00::/7. In normal private-network configurations, ULAs commonly begin with fd. They can be used for communication between devices on the same LAN without requiring globally routable IPv6 addresses.
- For example, two devices on a local network can use fd12:3456:789a::10 and fd12:3456:789a::20 to communicate, provided their addresses and network configuration are compatible.

➢ **Requirements**

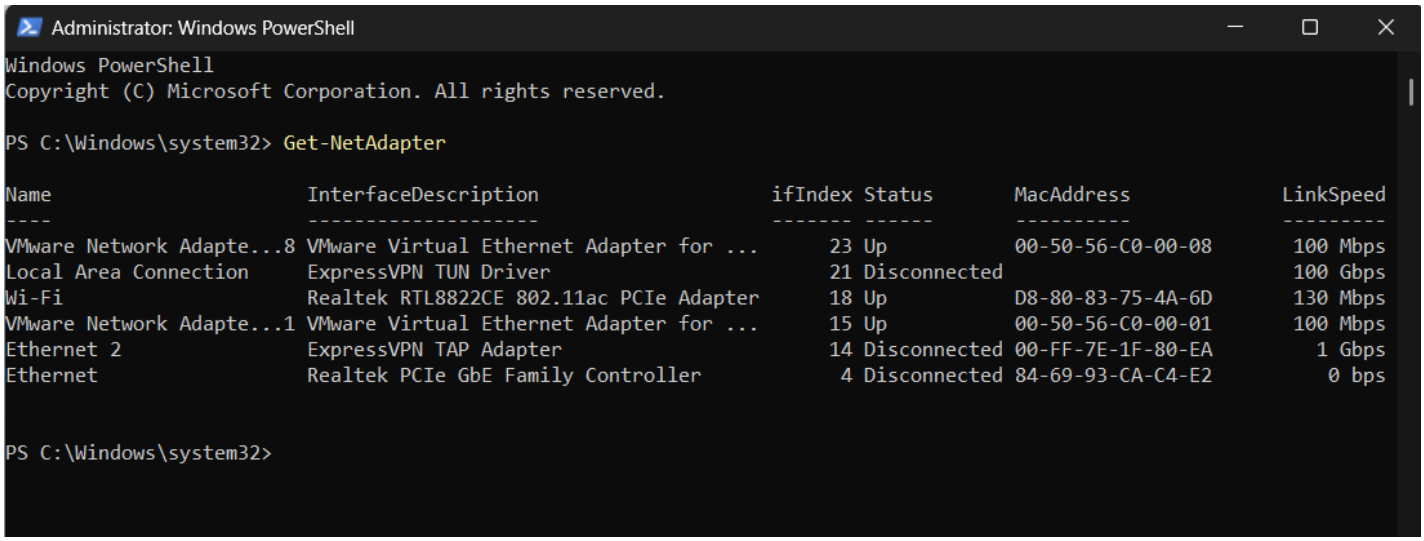
- A Windows or Linux computer
- Another device connected to the same LAN
- Administrator or root privileges
- IPv6 support enabled on both devices
- Command Prompt, PowerShell, or Terminal

➢ **Procedure for Windows**

➢ **Step 1: Identify the Network Adapter**

➢ Open PowerShell as Administrator and execute:
Get-NetAdapter

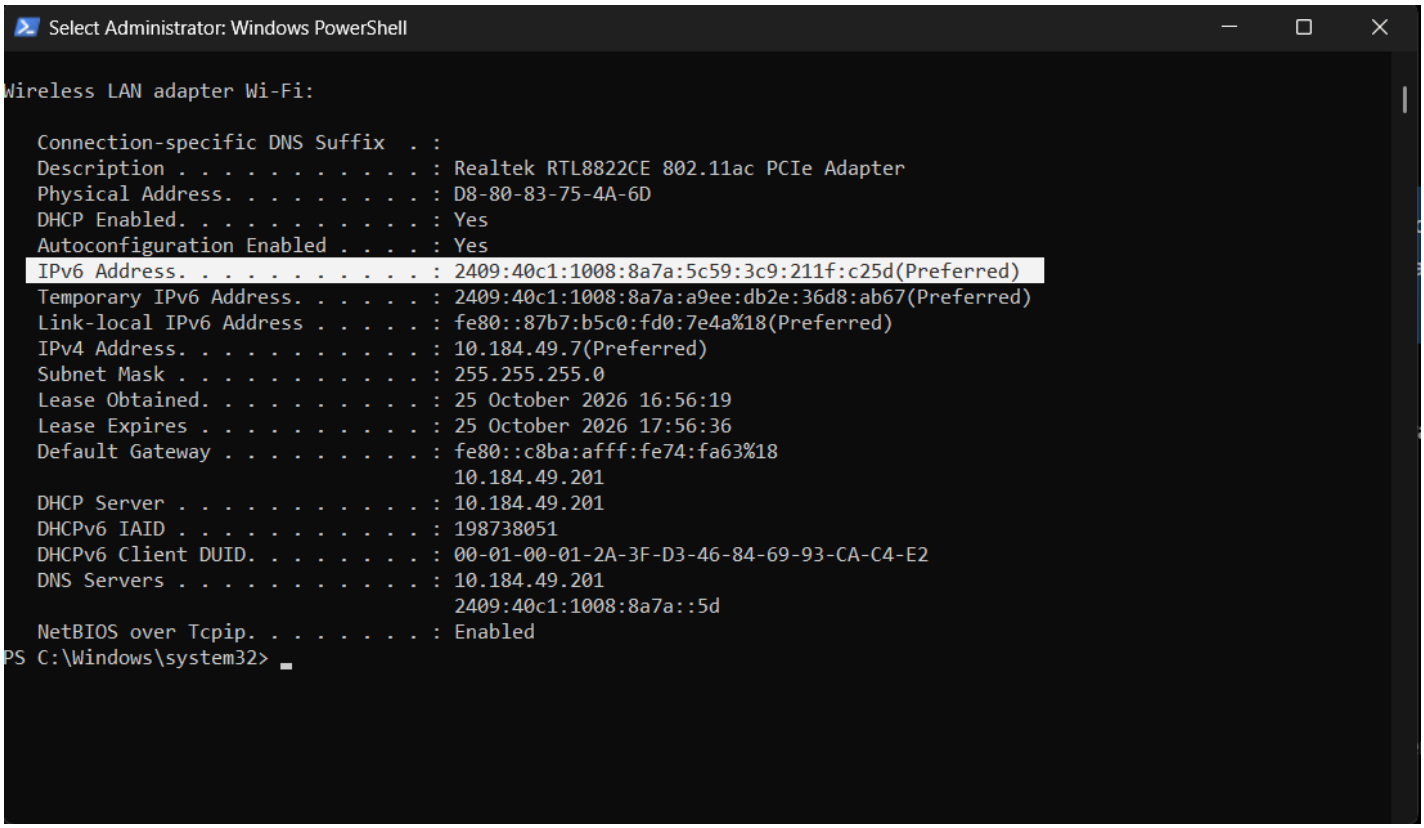
➢ Identify the active network adapter, such as Wi-Fi or Ethernet.



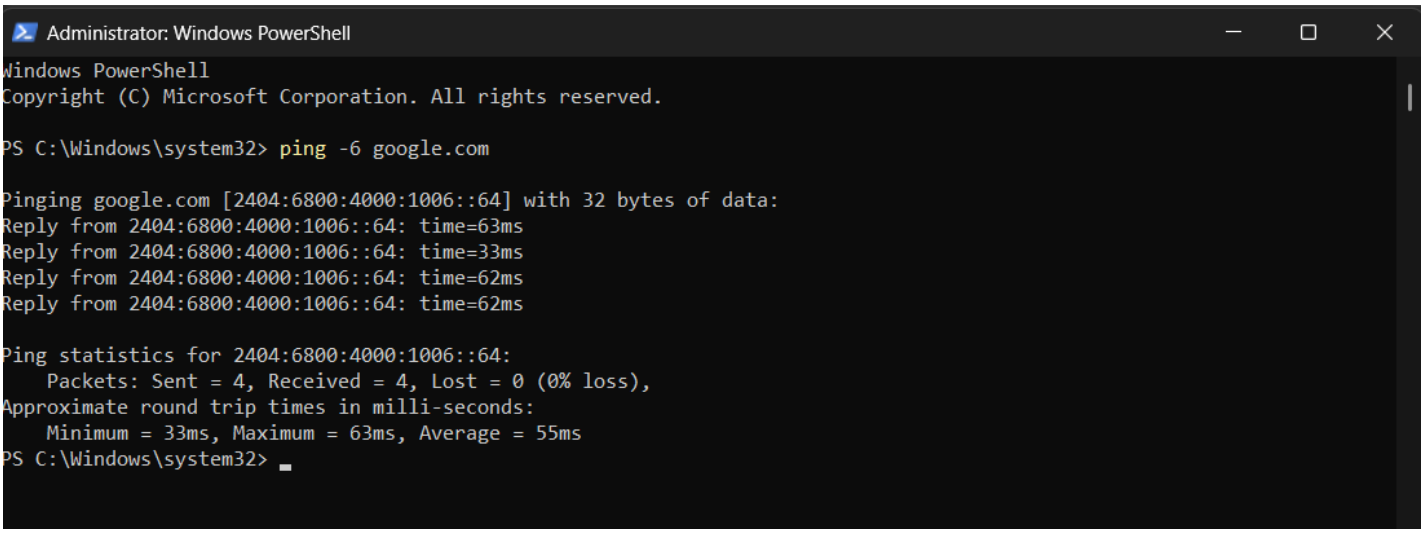
- **Step 2: Configure a Unique Local IPv6 Address**
- For a controlled lab network, assign a unique IPv6 address to each device. For example, if the network is configured to use the prefix fd12:3456:789a::/64, the two devices can use:
  - **Device 1: fd12:3456:789a::10/64**
  - **Device 2: fd12:3456:789a::20/64**
- On the first Windows device, execute the following command after replacing Wi-Fi with the actual adapter name:
  - **New-NetIPAddress -InterfaceAlias "Wi-Fi" -AddressFamily IPv6 -IPAddress "fd12:3456:789a::10" -PrefixLength 64**
- On the second device, configure the address fd12:3456:789a::20 with the same prefix length.
- **Step 3: Verify the IPv6 Address**
- On each device, execute:
  ipconfig
- For detailed information, execute:

ipconfig /all

- Locate the active network adapter and verify that the configured IPv6 address is displayed.



- **Step 4: Test Connectivity**
- From the first device, ping the IPv6 address of the second device:
  ping -6 fd12:3456:789a::20
- If the configuration is correct and ICMPv6 traffic is permitted, replies should be received from the second device.



- **Procedure for Linux**
- **Step 1: Identify the Network Interface**
- Open the Terminal and execute:
  ip link show
- Identify the active interface, such as eth0, ens33, or wlan0.
- **Step 2: Configure the Unique Local IPv6 Address**
- On the first device, execute:
  sudo ip -6 addr add fd12:3456:789a::10/64 dev eth0
- On the second device, execute:
  sudo ip -6 addr add fd12:3456:789a::20/64 dev eth0
- Replace eth0 with the actual network interface name on each computer.
- **Step 3: Verify the Address**
- Execute:
  ip -6 addr show
- Check that the assigned fd IPv6 address appears under the correct interface.
- **Step 4: Test Connectivity**

- On the first Linux device, execute:
  ping -6 fd12:3456:789a::20
- On Linux systems that support the ping6 command, the alternative is:
  ping6 fd12:3456:789a::20
- Successful replies indicate that IPv6 communication is working between the two devices.
- **Observation**

Device	Example Unique Local IPv6 Address	Prefix Length
Device 1	`fd12:3456:789a::10`	64
Device 2	`fd12:3456:789a::20`	64

- Both example addresses belong to the same /64 subnet and are different from each other. Actual observations should be recorded from the devices used during the practical.
- **Result**
- A Unique Local IPv6 address beginning with fd was configured on a network device. The address was verified using the operating system's IPv6 configuration command, and connectivity was tested by pinging another device on the same LAN using ping -6 or ping6.

❖ **3. Set up a temporary IPv6 address on your system and check if it changes after a reboot or network disconnect/reconnect. Document the steps and the observed address behavior in a text file.**

➢ **Aim**

- To observe the temporary IPv6 address assigned to a Windows system and check whether it changes after disconnecting and reconnecting to the network or restarting the system.

➢ **Introduction**

- IPv6 (Internet Protocol Version 6) is used to identify devices on a network. A temporary IPv6 address is a privacy feature that helps reduce the tracking of a device by using an address that can change over time.
- Windows can use temporary IPv6 addresses for outgoing connections when the relevant privacy settings are enabled. However, the address does not necessarily change every time the system reconnects to Wi-Fi or restarts.

➢ **Requirements**

- Windows laptop or computer
- Wi-Fi Internet connection
- Command Prompt or PowerShell
- Notepad for recording observations

➢ **Procedure for Windows**

1. Connected the laptop to a Wi-Fi network.
2. Opened Command Prompt.

3. Executed the command ipconfig /all.
4. Located the Wireless LAN adapter Wi-Fi section.
5. Recorded the IPv6 Address, Temporary IPv6 Address, and Link-local IPv6 Address.
6. Disconnected and reconnected the Wi-Fi network.
7. Executed ipconfig /all again and compared the IPv6 addresses.
8. Restarted the computer and checked the addresses again.
9. Recorded the observed results in a text file.

- **Initial IPv6 Address Configuration**
- The following details were observed on the Windows system:

Parameter	**Observed Value**
Network Adapter	Realtek RTL8822CE 802.11ac PCIe Adapter
DHCP Enabled	Yes
Autoconfiguration Enabled	Yes
IPv6 Address	`2409:40c1:1008:8a7a:5c59:3c9:211f:c25d`
Temporary IPv6 Address	`2409:40c1:1008:8a7a:a9ee:db2e:36d8:ab67`
Link-local IPv6 Address	`fe80::87b7:b5c0:fd0:7e4a%18`
IPv4 Address	`10.184.49.7`

Subnet Mask	255.255.255.0
Default Gateway (IPv4)	10.184.49.201

- **Observation**
- **A. After Wi-Fi Disconnect and Reconnect**
  After reconnecting the Wi-Fi network, the IPv6 address was checked again using **ipconfig /all.**
  Observed result: The address appeared to remain the same after reconnecting to Wi-Fi.
- **B. After System Restart**
  After restarting the computer, the IPv6 address must be checked again and compared with the initial values.
  Observed result: To be recorded after the restart test.

Test	Normal IPv6 Address	Temporary IPv6 Address
Initial observation	2409:40c1:1008:8a7a:5c59:3c9:211f:c25d	2409:40c1:1008:8a7a:a9ee:db2e:36d8:ab67
After Wi-Fi reconnection	Same as initial	Same as initial
After system restart	Same as initial	Same as initial

- **Result**
- The IPv6 configuration of the Windows system was successfully examined using the ipconfig /all command. The normal IPv6 address, temporary IPv6 address, and link-local IPv6 address were identified.
- During testing, both the normal IPv6 address and the temporary IPv6 address remained unchanged after Wi-Fi reconnection and system restart.

❖ **4. Simulate a Zomato-style search: Write a small script (any language) that stores an array of restaurant names and their IPv6 addresses, then allows the user to input a partial name and returns the matching restaurant's IPv6 address.**

- **Aim**
  - To write a Python script that stores restaurant names and their IPv6 addresses in an array-like data structure and allows users to search for a restaurant using a partial name to retrieve its corresponding IPv6 address.

- **Introduction**
  - Zomato is an online food discovery and ordering platform that allows users to search for restaurants by name. A similar search feature can be simulated using a simple Python program.
  - In this program, a collection of restaurant names and their example IPv6 addresses is stored in a list of dictionaries. The user enters a partial

restaurant name, and the program searches for matching names and displays the corresponding IPv6 addresses.

- **Python Program**

```
# Zomato-Style Restaurant Search Using IPv6

restaurants = [
    {
        "name": "Pizza Palace",
        "ipv6": "2001:db8:1::10"
    },
    {
        "name": "Burger House",
        "ipv6": "2001:db8:1::20"
    },
    {
        "name": "Spice Garden",
        "ipv6": "2001:db8:1::30"
    },
    {
        "name": "Royal Biryani",
        "ipv6": "2001:db8:1::40"
    },
    {
        "name": "Tasty Dosa",
        "ipv6": "2001:db8:1::50"
    }
```

```
]

search_name = input("Enter restaurant name or part of name: ")
found = False

for restaurant in restaurants:
    if search_name.lower() in restaurant["name"].lower():
        print("Restaurant:", restaurant["name"])
        print("IPv6 Address:", restaurant["ipv6"])
        print("-------------------------")
        found = True

if not found:
    print("No matching restaurant found.")
```

- **Sample Output**
- **Example 1: Matching restaurant found**

```
Enter restaurant name or part of name: pizza

Restaurant: Pizza Palace
IPv6 Address: 2001:db8:1::10
-------------------------
```

- **Example 2: Partial name search**

```
Enter restaurant name or part of name: bir

Restaurant: Royal Biryani
IPv6 Address: 2001:db8:1::40
```

- **Example 3: No matching restaurant found**

  Enter restaurant name or part of name: subway

  No matching restaurant found.

- **Explanation of the Program**

1. **Restaurant Data:** A list of dictionaries stores each restaurant's name and its corresponding IPv6 address.
2. **User Input:** The input() function accepts a full or partial restaurant name from the user.
3. **Search Operation:** The for loop checks each restaurant in the list.
4. **Partial Matching:** The in operator checks whether the entered text appears within a restaurant's name.
5. **Case-Insensitive Search:** The lower() function allows searches using uppercase or lowercase letters.
6. **Display Results:** If a match is found, the restaurant's name and IPv6 address are displayed.
7. **No Match Handling:** If no restaurant matches the entered text, the program displays an appropriate message.

- **Result**
- The Python program was designed to simulate a Zomato-style restaurant search. It accepts a partial restaurant name, searches the stored restaurant records, and displays the matching restaurant's IPv6 address. It also handles cases where no matching restaurant is found.

❖ **5. Use ChatGPT or Copilot to generate a sample IPv6 subnetting scenario (with at least 3 subnets), and then manually assign a valid IPv6 address from one of those subnets to your system. Paste the AI-generated scenario and your configuration steps in a text file.**

➢ **Aim**

- To generate an IPv6 subnetting scenario using ChatGPT, create at least three subnets, and assign a valid IPv6 address from one of the subnets to a Windows system.

➢ **Introduction**

- IPv6 (Internet Protocol Version 6) is a network addressing protocol that uses 128-bit addresses to identify devices on a network. IPv6 subnetting divides a larger network into smaller logical networks, making network management easier.
- ChatGPT can be used to generate an IPv6 subnetting plan. The selected IPv6 address can then be configured on a computer using Windows PowerShell.

➢ **AI-Generated IPv6 Subnetting Scenario**
➢ **Given IPv6 Network Prefix:** fd12:3456:789a::/48
➢ The network is divided into three /64 subnets for different departments.

No.	Department	IPv6 Subnet
1	Administration	`fd12:3456:789a:1::/64`
2	Sales	`fd12:3456:789a:2::/64`

3	Technical Support	fd12:3456:789a:3::/64

- The given network uses a Unique Local Address (ULA) prefix, which is intended for private network communication.

- **Selected IPv6 Address**
- For this practical, the Administration subnet is selected.
  - **Subnet Prefix:** fd12:3456:789a:1::/64
  - **IPv6 Address:** fd12:3456:789a:1::10
  - **Prefix Length:** 64
  - **Network Interface:** Wi-Fi
- The selected address belongs to the Administration subnet because it falls within the specified /64 prefix.

- **Configuration Steps in Windows**
  1. Open Windows PowerShell as Administrator.
  2. Check the available network interfaces using the command:
     Get-NetAdapter
  3. Confirm the name of the Wi-Fi interface.
  4. Assign the selected IPv6 address using the following command:
     New-NetIPAddress -InterfaceAlias "Wi-Fi" -IPAddress "fd12:3456:789a:1::10" -PrefixLength 64 -AddressFamily IPv6
  5. Verify the IPv6 address using:
     Get-NetIPAddress -InterfaceAlias "Wi-Fi" -AddressFamily IPv6
  6. Check whether the assigned address appears in the command output.

- **Observation**
  - The AI-generated subnetting plan contains three IPv6 subnets for Administration, Sales, and Technical Support. The sample address fd12:3456:789a:1::10 was selected from the Administration subnet.
  - The address assignment must be verified using PowerShell. The actual result depends on whether the configuration command executes successfully on the system.

- **Result**
- An IPv6 subnetting scenario with three /64 subnets was prepared using ChatGPT. A valid sample IPv6 address was selected from one subnet, and the Windows commands required to assign and verify the address were identified.